

Automated Cell Segmentation in Microscopy Images Using U-Net for Breast Cancer Studies

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Abstract—This study presents an automated approach for cell segmentation in microscopy images using U-Net architecture, specifically designed for breast cancer research. We utilized a custom dataset from Georgia Institute of Technology comprising microscopy images with three distinct channels: bright-field, nuclear fluorescence, and grid structure. Two training scenarios were investigated: one using single-experiment data and another incorporating data from multiple experiments. While the first scenario achieved higher performance metrics on familiar data, the second scenario demonstrated superior generalization capabilities across different experimental conditions, despite a slight decrease in overall performance. The model successfully identified and segmented cell nuclei, though challenges remained in separating overlapping cells. Our results indicate that diverse training data is crucial for developing robust segmentation models in biological imaging applications, particularly when dealing with varying experimental conditions and imaging artifacts. This work contributes to the advancement of automated cell tracking tools, facilitating large-scale studies of cancer cell mobility and metastasis.

I. INTRODUCTION

Cancer remains one of the leading causes of mortality worldwide, responsible for nearly 10 million deaths annually [1]. Despite significant advances in detection and treatment, it continues to present complex challenges in healthcare. The primary cause of cancer-related mortality is metastasis—the process by which cancer cells detach from the primary tumor, migrate through the body, invade healthy tissues, and establish secondary tumors [2]. Understanding the mechanisms behind metastasis is crucial for developing targeted therapies that could inhibit this process, improve survival rates, and enhance the quality of life for cancer patients [3].

Cancer cell movement has traditionally been attributed to myosin proteins, which generate contractile forces similar to those in muscle cells [4]. However, recent research has revealed an alternative pathway involving dynein proteins, which function as molecular motors enabling cancer cell migration even when myosin activity is inhibited. Remarkably, cancer cells can continue migrating if either protein remains functional, but their

movement stops entirely when both are inhibited [3]. This discovery of dual mechanisms of movement underscores the adaptability of cancer cells and highlights the critical need for accurate tracking to better understand and potentially disrupt metastatic behavior.

Accurate tracking of cancer cells is fundamental to advancing cancer research, particularly in evaluating the effects of various treatments on cell movement. Traditional image processing techniques, while valuable, face significant limitations including sensitivity to noise, poor generalization to complex data, and dependence on manual parameter tuning. In contrast, deep learning pipelines offer a more robust alternative, with the ability to handle large datasets, adapt to diverse imaging conditions via transfer learning, and uncover subtle patterns in data that may be imperceptible to classical methods, enabling more precise tracking of cancer cell movements even in challenging scenarios, such as low-contrast or noisy images.

AI-driven cancer cell tracking methods generally fall into two main categories. The first utilizes object detection and tracking models, such as YOLO (You Only Look Once) [5], to directly identify and follow individual cancer cells across sequential frames. The second emphasizes semantic segmentation, where binary masks delineating cell nuclei are generated. These masks serve as the foundation for subsequent tracking, either through classical image processing techniques or advanced AI-based pipelines. In this study, we focus on the segmentation approach, leveraging pre-trained U-Net models [6], which have been widely and successfully used for biomedical image segmentation.

We hypothesize that leveraging AI-based segmentation pipelines will significantly enhance the accuracy and reliability of cancer cell tracking. This advancement would enable researchers to conduct more experiments under diverse conditions, facilitating the evaluation of treatment effects on cancer cell mobility. By establishing a robust framework for automated segmentation and tracking, this work seeks to advance the tools available for studying metastasis, ultimately contributing to a deeper understanding of cancer progression and supporting the development of more effective therapeutic strategies.

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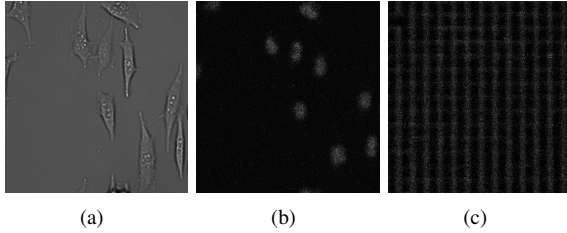


Fig. 1: Representative images from the dataset illustrating the three distinct channels. (a) bright-field image showing the normal view of cells during migration, (b) fluorescence image highlighting nuclear activity, and (c) grid structure used as a reference for tracking cell movements.

II. MATERIALS AND METHODS

A. Dataset

The dataset used in this study was collected at the Georgia Institute of Technology to investigate cell locomotion and dynamics in breast cancer tissues under varying conditions. This custom dataset comprises microscopy images obtained from experiments designed to explore the effects of Blebbistatin, a treatment known to inhibit actomyosin contractility, as well as control conditions with no treatment. For each experiment, imaging was conducted at five distinct positions within the experimental setup. Each captured frame has a resolution of 2018×2018 pixels, providing high spatial detail necessary for the analysis of cellular behavior. The dataset includes three distinct imaging channels: one channel captures the grid-like structure on which the cells migrate, another provides standard bright-field images of the cells as they move, and the third highlights nuclear fluorescence, which is crucial for understanding nuclear dynamics during cell migration. A sample of the images representing these channels has been included in Figure 1 for reference. The dataset facilitates the use of artificial intelligence techniques to study cell movements in response to various interventions. Specifically, it supports the examination of how cells interact with their microenvironment, both in the presence and absence of Blebbistatin, thereby offering valuable insights into the mechanisms of cellular motility under diverse conditions.

B. Preprocessing

In this study, the grid channel was not used, and preprocessing focused on the nuclear and regular image channels. Both the nuclear image and the bright-field image channels were resized to a resolution of 2048×2048 pixels, as this resolution was more convenient for subsequent processing and analysis. For the nuclear image channel, a fixed threshold value of 20 was applied

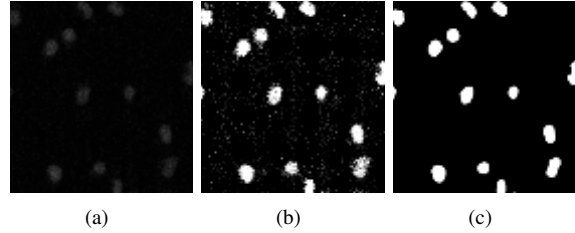


Fig. 2: Steps involved in preprocessing to create masks from the nuclear channel. (a) The raw nuclear image, (b) the image after applying a threshold of 20, and (c) the final mask obtained by applying median filtering with a kernel size of 9×9 to remove small artifacts.

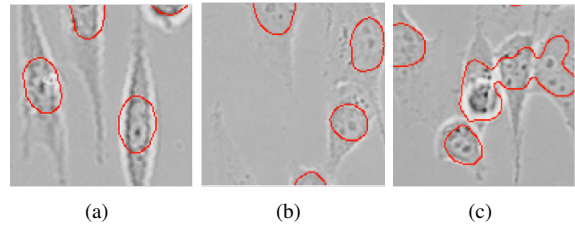


Fig. 3: Each patch displays the original grayscale bright-field microscopy image overlaid with red contours indicating cell nuclei boundaries from ground truth masks. Panels (a) and (b) demonstrate accurate cell segmentation, while panel (c) illustrates a limitation where adjacent cells were incorrectly segmented as one object during mask generation.

to create masks by isolating regions of interest. To further refine these masks and remove small artifacts resembling salt-and-pepper noise, median filtering with a kernel size of 9×9 was applied. The steps involved in preprocessing for creating masks are represented in Figure 2. The regular image channel underwent only resizing to 2048×2048 pixels without any additional processing. Finally, both the processed nuclear and bright-field images were divided into smaller patches of size 128×128 pixels to prepare them for use in the training pipeline, facilitating efficient processing and compatibility with the model input requirements. Some of the patches of the data, created from the regular and nuclear channels, are demonstrated in Figure 3.

C. Pipeline

A U-Net model was employed to segment nuclei in microscopy images. U-Net is a convolutional neural network architecture widely used for biomedical image segmentation. It features an encoder-decoder structure, where the encoder captures contextual information through successive downsampling, and the decoder re-

constructs the segmentation map through upsampling. Skip connections between corresponding encoder and decoder layers allow the network to combine high-level features with spatial details, making it particularly effective for segmenting small and intricate structures. Figure 4 provides a visual overview of the U-Net architecture.

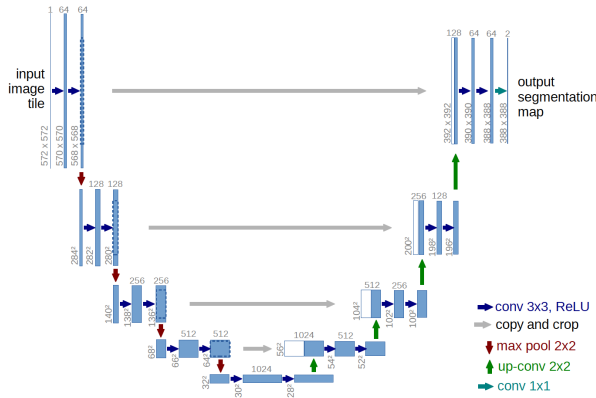


Fig. 4: The structure of the U-Net architecture presented in the original paper. This architecture enables precise segmentation of small and intricate structures by integrating spatial details with high-level contextual information. [6]

In this study, patches of size 128×128 pixels from the regular image channel were used as input to the U-Net model. The model predicted a binary mask for each patch, with 0 representing the background and 1 representing the nuclei of the cells. A pretrained U-Net model with a ResNet34 backbone, initialized with weights learned on the ImageNet dataset, was utilized to leverage transfer learning. This approach enhanced feature extraction and reduced training time.

The model was trained with a batch size of 2048, utilizing a learning rate of 0.001 and the Adam optimizer to ensure efficient and stable optimization. Cross-entropy was employed as the loss function to minimize prediction errors, while the Dice metric was used during validation to assess the model's performance. The model was fine-tuned over 20 epochs, achieving accurate segmentation results without data augmentation or postprocessing.

III. RESULTS

Two training scenarios were designed to evaluate and improve the segmentation performance of the model, with qualitative and quantitative analyses supported by visual results. Early stopping was used in both scenarios to select the model with the best validation loss, ensuring optimal performance.

In the first training scenario, only data from experiment 1 was used. Out of the 5 available positions for the two types of cells, 8 movies were allocated for training,

1 for validation, and 1 for testing. The model performed well on the testing data from experiment 1, as evidenced by loss curves (Figure 5(a)), the training and validation Dice metric trends (Figure 5(b)), and qualitative results (Figure 6). Visual segmentation results showed accurate identification of cells in most cases, but challenges arose in regions with overlapping cells. In these regions, overlapping cells were often merged and treated as a single entity, reducing segmentation accuracy. When the model was tested on completely unseen data from experiment 2, its performance degraded significantly. Visual inspection (Figure 7) revealed that many grid-like structures, which are artifacts in the dataset, were misclassified as cells. This highlighted the model's inability to generalize to data with differing characteristics.

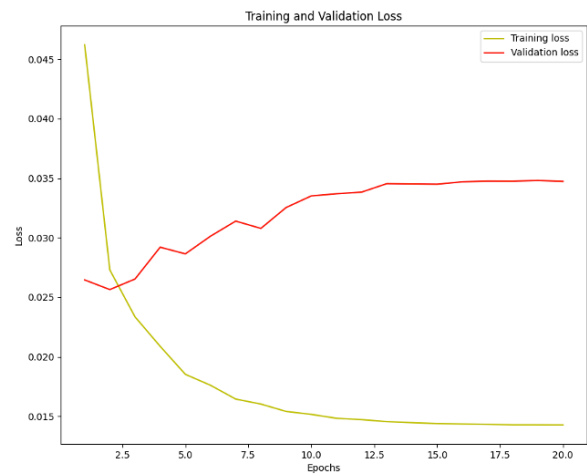
To address these limitations, a second training scenario was implemented. This approach included data from both experiment 1 and experiment 2, with one position from each experiment and each type of cell used for training (4 movies), two for validation, and the remaining movies for testing. Early stopping was again applied to ensure the best-performing model was selected. While there was a slight decrease in overall performance, as evidenced by the Dice metric trends and loss curves (Figure 8), the model's ability to generalize improved significantly. Qualitative results (Figure 9) showed that grid-like structures were no longer misclassified as cells, and the model exhibited more reliable segmentation on unseen data from both experiments. This improvement in generalizability outweighed the minor reduction in overall performance, demonstrating the model's robustness across different datasets.

Quantitative analysis further supports these observations. The Dice metric and loss curves for training and validation revealed a significant gap between training and validation metrics in the first scenario, particularly for unseen data, indicating poor generalization. In the second scenario, this gap was reduced considerably, demonstrating better generalization across experiments. Despite the slight reduction in overall segmentation performance observed in the second training scenario, its improved generalizability to unseen data makes it a more reliable approach. Some challenges remain, such as reliably separating overlapping cells and addressing variations in image quality, but these issues were less pronounced in the second scenario compared to the first.

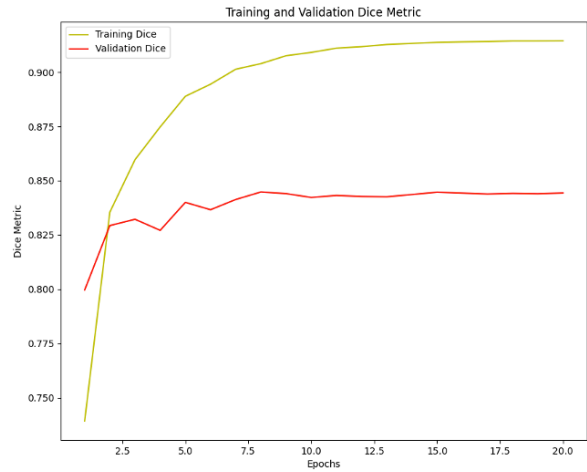
IV. DISCUSSION

A. U-net for semantic cell segmentation

The results of this study demonstrate the effectiveness of U-Net architecture for automated cell segmentation in microscopy images, while also highlighting important considerations for developing robust segmentation models. The comparison between our two training scenarios



(a)



(b)

Fig. 5: Loss and Dice metric trends for the first training scenario. (a) Training and validation loss over epochs, demonstrating convergence and the selection of the best-performing model using early stopping. (b) Training and validation Dice metric over epochs, highlighting the model’s strong performance on the validation set within experiment 1.

reveals crucial insights about model generalization and the importance of diverse training data in biological image analysis.

Our first training scenario, while achieving high-performance metrics on data from experiment 1, showed limited generalization capability when applied to experiment 2 data. This observation aligns with previous findings in biomedical image analysis, where models trained on homogeneous datasets often struggle with domain adaptation [7]. The misclassification of grid-

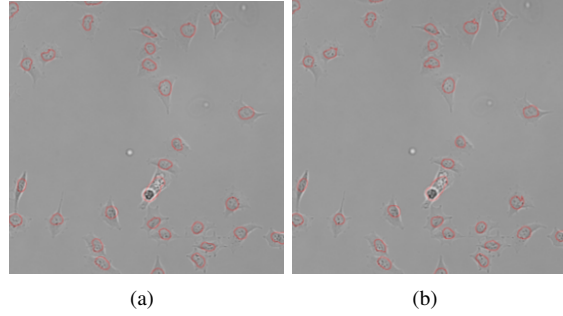


Fig. 6: Visual testing of the segmentation pipeline for the first training scenario on experiment 1. (a) Ground truth, and (b) predicted segmentation. While the model performs well overall, the performance is slightly degraded in regions with overlapping cells, where they are often merged into a single entity.

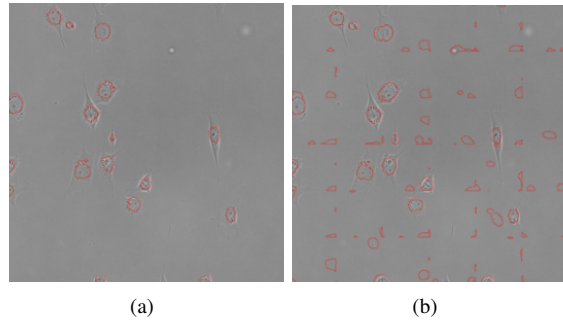


Fig. 7: Visual testing of the segmentation pipeline for the first training scenario on experiment 2. (a) Ground truth, and (b) predicted segmentation. The model struggles to generalize to unseen data from experiment 2, misclassifying grid-like structures as cells.

like structures as cells in experiment 2 particularly emphasizes the importance of exposure to diverse imaging conditions during training.

The second training scenario, incorporating data from both experiments, demonstrated improved generalization despite a slight decrease in overall performance metrics. This trade-off between specialized performance and generalization capability reflects a common challenge in deep learning applications for biomedical image analysis [8]. The reduced misclassification of grid-like structures and more reliable cell detection across different experimental conditions suggests that diverse training data can help models develop more robust feature representations.

B. Limitations

Several limitations of our approach warrant discussion. First, the current implementation struggles with

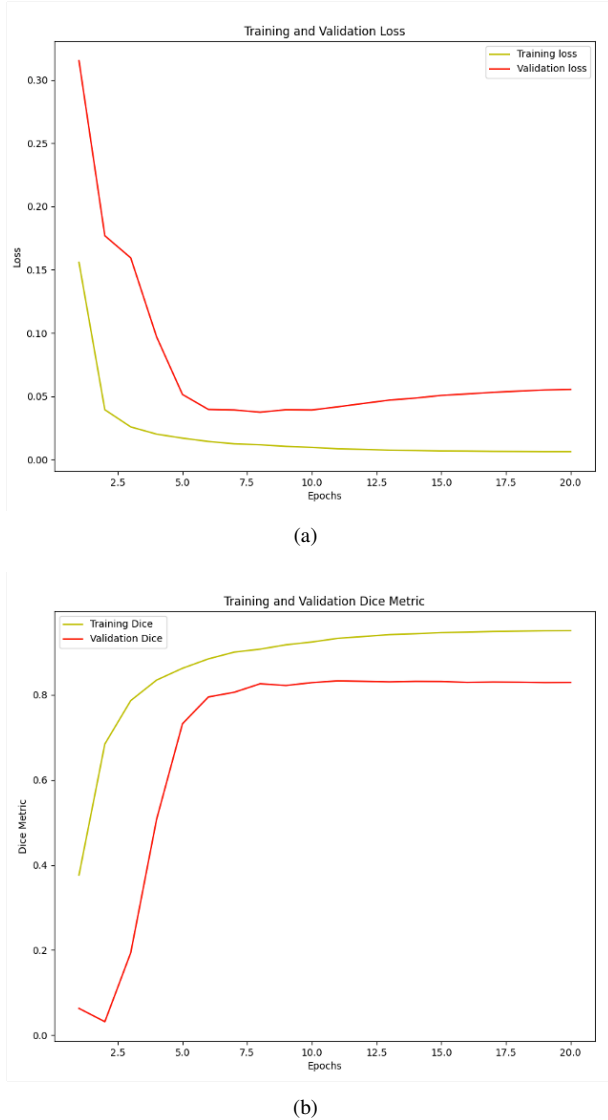


Fig. 8: Loss and Dice metric trends for the second training scenario. (a) Training and validation loss over epochs, showing a slight degradation in loss compared to the first training scenario. (b) Training and validation Dice metric over epochs, reflecting a minor decrease in performance. However, this training scenario significantly improved the model’s generalizability to unseen data from both experiments.

accurately segmenting overlapping cells, a common challenge in cell segmentation tasks [9]. This limitation could potentially impact tracking accuracy in densely populated regions. Second, while our model performs well on the available experimental conditions, its generalization capability to different microscopy settings or cell types remains untested. Third, the current pipeline requires manual threshold selection for preprocessing,

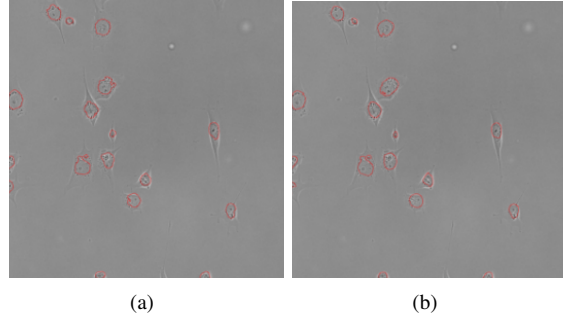


Fig. 9: Visual testing of the segmentation pipeline for the second training scenario. (a) Ground truth, and (b) predicted segmentation. The model demonstrates improved generalizability, correctly identifying cells and reducing misclassification of grid-like structures compared to the first training scenario.

which could introduce variability across different experimental setups. More sophisticated approaches, such as optical flow algorithms, could potentially create more robust and reliable ground truth masks by incorporating temporal information and cell movement patterns, thereby improving the overall segmentation accuracy.

C. Future works

Future work should focus on several key areas. First, incorporating instance segmentation capabilities could address the challenge of separating overlapping cells. Second, exploring self-supervised learning approaches might reduce the need for extensive manually labeled training data. Third, developing adaptive preprocessing techniques could eliminate the need for manual threshold selection. Fourth, the availability of a dedicated grid channel in our dataset presents an opportunity to improve the model’s ability to distinguish between cellular structures and grid artifacts - this could be achieved through multi-channel input processing or using the grid channel for explicit artifact removal. Finally, validating the pipeline across a broader range of experimental conditions and cell types would further establish its utility for general biological research applications.

V. CONCLUSION

Our study demonstrates the effectiveness of deep learning-based approaches, particularly U-Net, for automated cell segmentation in microscopy images while highlighting the critical trade-off between specialized performance and generalization capability. The improved generalization achieved through diverse training data, despite a minor reduction in overall performance metrics, establishes a practical framework for developing robust

cell segmentation tools. This advancement enables more efficient analysis of cancer cell behavior under various experimental conditions, supporting research into cancer metastasis mechanisms. While challenges remain, particularly in separating overlapping cells and handling imaging artifacts, our findings suggest that incorporating multi-channel information and advanced preprocessing techniques could further enhance segmentation accuracy. The demonstrated success of our approach, combined with proposed future improvements, positions this work as a valuable contribution to both technical advancement in biomedical image analysis and practical applications in cancer research.

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